INCH-POUND

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SUPERSEDING
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DETAIL SPECIFICATION

RINGS, DEE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers dee rings for use on various equipage items (see 6.1).

1.2 Classification. The dee rings will be of the configurations shown by letter designations on drawings 4-1-454 or 4-1-455, as specified (see 6.2) and will be of the following classes and sizes for the configuration shown. Configuration X will be style 1 (without roller) or style 2 (with roller), as specified (see 6.2).

1.2.1 Classes. Classes of dee rings will be as follows (see 6.1), as specified (see 6.2):

Class 1- Brass
Class 2- Steel wire
Class 3- Malleable iron
Class 4- Nickel-silver alloy
Class 5- Molding plastic (nylon) rigid

1.2.2 Configurations. The configurations of the dee rings are shown by letter designations on drawings 4-1-454 and 4-1-455, as specified (see 6.2).

1.2.3 Sizes. Configurations A, D, K, L, W, and X are available in various sizes as shown on drawing 4-1-454, as specified (see 6.2).

Comments, suggestions, or questions on this document should be addressed to Defense Logistics Agency Aviation VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616, or e-mailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST database at https://assist.dla.mil/.

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1.2.4 Styles. The styles (configuration X only) are as follows: Style 1 for dee rings without rollers, and style 2 for dee rings with rollers (brass and steel only), as shown on drawing 4-1-454, as specified (see 6.2).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specification and standard. The following specification and standard form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATION

MIL-F-495	- Finish, Chemical, Black, for Copper Alloys

DEPARTMENT OF DEFENSE STANDARD

MIL-STD-1916	- DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.2.2 Other Government drawings. The following other Government drawings form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DRAWINGS

U.S. ARMY NATICK SOLDIER RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

4-1-454	- Rings, Dee, SYM A, C, D, K, L, M, U, W, and X
4-1-455	- Rings, Dee, SYM B, E, F, G, H, J, R, T, and V

(Copies of these drawings are available from the Technical Director, U.S. Army Combat Capabilities Development Command Soldier Center, Attn: FCDD-SCE-E, 15 General Greene Avenue, Natick, MA 01760-5056.)

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ H1331	- Zero Acceptance Number Sampling PlansSampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at http://www.asq.org.)

ASTM INTERNATIONAL

ASTM A47/A47M	- Standard Specification for Ferritic Malleable Iron Castings
ASTM A853	- Standard Specification for Steel Wire, Carbon, for General Use
ASTM B21/B21M	- Standard Specification for Naval Brass Rod, Bar, and Shapes
ASTM B36/B36M	- Standard Specification for Brass Plate, Sheet, Strip, and Rolled Bar
ASTM B584	- Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM B633	- Standard Specification for Electrodeposited Coatings of Zinc on Iron and Steel
ASTM D4066	- Standard Classification System for Nylon Injection and Extrusion Materials (PA)
ASTM E1282	- Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores, and Related Materials

(Copies of these documents are available at http://www.astm.org/.)

MASTER PAINTERS INSTITUTE (MPI)

MPI #94	- Alkyd, Exterior, Semi-Gloss (MPI Gloss Level 5)

(Copies of this document are available at http://www.paintinfo.com/.)

SAE INTERNATIONAL

SAE AMS 2460	- Plating, Chromium
SAE AMS-QQ-P-416	- Plating, Cadmium (Electrodeposited)
SAE-AMS-STD-595/27038	- Miscellaneous, Semigloss

(Copies of these documents are available at http://www.sae.org/.)

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2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Materials. The materials specified shall conform to applicable specifications and requirements specified herein. Commercial tolerances apply to wire diameters.

3.2.1 Copper alloy casting. The copper alloy (brass) castings shall conform to copper alloy UNS no. C85700 in accordance with ASTM B584.

3.2.2 Brass sheet or strip. The brass sheet or strip shall conform to copper alloy UNS no. C26000 or C26800, rolled temper half hard in accordance with ASTM B36/B36M.

3.2.3 Wire, copper alloy. The copper alloy (brass) wire shall conform to copper alloy UNS no. C46200 or C46400, temper half hard or light annealed in accordance with ASTM B21/B21M.

3.2.4 Steel wire. Steel wire shall conform to AISI steel grade number 1006, 1008, or 1010, bare, annealed-in-process in accordance with ASTM A853.

3.2.5 Malleable iron. Malleable iron castings shall be black, ungalvanized, with a yield strength of 35,000 pounds per square inch (psi) and demonstrate an elongation of 18 percent minimum with a 2-inch test sample tested in accordance with ASTM A47/A47M.

3.2.6 Nickel-silver alloy. Nickel-silver alloy castings shall conform to table I for chemical composition when tested as specified in 4.3.3.

TABLE I. Chemical composition, copper-nickel-zinc (nickel silver) alloy.

Material	Percent
Copper	55.0 to 64.0
Nickel	18.0 (min)
Zinc	Remainder
Iron	0.35 (max)
Other	0.25 (max)

3.2.7 Enamel. The enamel shall be semi-gloss enamel alkyd conforming to MPI #94.

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3.2.8 Plastic, nylon. The plastic dee rings shall be molded from virgin nylon conforming to designation PA0181 of ASTM D4066. Clean unburned plastic material in the form of imperfect parts, sprues, runners, or other scrap of the same composition as the virgin material, and produced in the molding or finishing operation, may be reground and mixed with the virgin material specified. When reground material is mixed with the virgin material, the reground material shall not exceed a level of 25 percent by weight of the blended mixture.

3.2.9 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Design and construction. The design and construction of the rings shall conform to drawings 4-1-454, and 4-1-455, and the requirements specified herein. All burrs, sharp edges, and slivers shall be removed before application of any finish.

3.3.1 Class 1 rings. At the option of the supplier, class 1 rings shall be cast of the brass material specified in 3.2.1 or formed of the brass wire specified in 3.2.3, except when only one material is shown on the applicable drawing. Available configurations are A, B, E, F, G, H, J, K, L, W, and X. After forming to shape and locating the joint as shown on the drawing, the joint shall be joined by butt resistance welding or brazing. The completed joints shall be smooth, round, and free of cracks, burns, or reduced areas. The rings shall be free of porosity, blowholes, warp, and clearly noticeable digs and gouges.

3.3.2 Class 2 rings. Class 2 rings shall be fabricated of the steel wire specified in 3.2.4. After forming to shape and locating the joint as shown on the drawing, the joint shall be butted and welded. Unless otherwise specified (see 6.2), configuration "A" rings shall be butted only. The welds shall be sound, not porous, and free from cracks, burns, or reduced areas. The welds shall be ground smooth and free of rough edges. The rings shall be free of porosity, blowholes, warp, and clearly noticeable digs and gouges. Available configurations are indicated on the drawings by the letters A, D, E, F, H, K, L, M, R, T, U, V, and X.

3.3.3 Class 3 rings. Class 3 rings shall be cast from the malleable iron specified in 3.2.5. Available configurations are A, B, E, F, G, J, W, and X. The rings shall be free of porosity, blowholes, warp, and clearly noticeable digs and gouges.

3.3.4 Class 4 rings. Class 4 rings shall be cast from the nickel-silver alloy specified in 3.2.6, conforming to configuration C. The rings shall be free of porosity, blowholes, warp, and clearly noticeable digs and gouges.

3.3.5 Class 5 rings. Class 5 rings shall be molded of the plastic specified in 3.2.8. The available configuration is X.

3.3.6 Rollers. The rollers required for configurations U, V, and X (style 2) shall be fabricated of the brass sheet or strip specified in 3.2.2 and to the dimensions shown on the drawings. After assembly to the straight section of the ring, the rollers shall rotate freely without binding and the joint of the roller along its entire length shall be butted evenly.

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3.4 Finishes. The finished rings shall be free from corrosion, dirt, discoloration, and areas of no finish. Finishes shall be as follows:

3.4.1 Black chemical. Black chemical finish shall conform to MIL-F-495.

3.4.2 Zinc-plate.

3.4.2.1 Zinc-plate without enameling. Zinc-plate without enameling shall conform to type II or type III, service condition (SC) 3 of ASTM B633.

3.4.2.2 Zinc-plate with enameling. Zinc-plate with enameling shall conform to type IV, SC3 of ASTM B633, followed by baking enamel. The enamel shall conform to 3.2.7.

3.4.3 Cadmium plate. Cadmium plate shall conform to class 1 or class 2, type I or type II of SAE AMS-QQ-P-416.

3.4.4 Chromium plate. Chromium plate shall conform to class 1 of SAE AMS 2460.

3.4.5 Class 1 rings. All class 1 rings, except configuration L, shall be finished as specified in 3.4.1.

3.4.5.1 Configuration L. Configuration L rings, except class 1 (1 inch x 7/8 inch), shall be given a bright, polished finish. Configuration L class 1 rings (1 inch x 7/8 inch) shall be given either a bright, polished finish or the chromium finish specified in 3.4.4, as specified (see 6.2). The chromium plate thickness shall be not less than 0.0002 inch.

3.4.6 Class 2 rings. All class 2 rings, except configurations L and X, shall be given a finish as specified in 3.4.2.2. The enamel coating shall level out to a uniform dry film thickness without orange peel, wrinkles, drops, streaks, or areas of no film.

3.4.6.1 Configuration L. At the option of the supplier, configuration L, class 2 dee rings shall be given either the finish specified in 3.4.2.1 or the class 2, type II finish specified in 3.4.3.

3.4.6.2 Configuration X. Configuration X, class 2 dee rings shall be given the type II or type III, SC3 finish specified in 3.4.2.1; or the class 1, type I finish specified in 3.4.3; as specified (see 6.2).

3.4.7 Class 3 rings. All class 3 dee rings shall be given the finish specified in 3.4.2.2. The enamel coating shall level out to a uniform dry film thickness without orange peel, wrinkles, drops, streaks, or areas of no film.

3.4.8 Class 4 rings. All class 4 dee rings shall be ground and polished to a bright finish.

3.4.9 Class 5 rings. The surfaces of the class 5 dee rings shall have a smooth and lusterless or semi-gloss finish. All surfaces shall be free of dirt, dust, and foreign matter inclusion. The finish shall be produced from a die that has the cavity surfaces finished in a vapor blast, water hone, or similar satin finish, and not result from the application of any protective

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coating lacquers or other materials. The resultant surface gloss shall not exceed that of SAE-AMS-STD-595/27038, semi-gloss black, by visual comparison.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When first article is required (see 6.2), the dee rings shall be examined for the defects specified in table II, the dimensions specified, and the tests of 4.3. The presence of any defect shall be cause for rejection of the first article.

TABLE II. End item visual defects.

Examination		Defect
Finish	General	Not type specified
		Evidence of corrosion
	Black chemical	Color not as specified
		Finish not clean and uniform
		Area of no finish or finish discolored
	Enamel	Area of no film or color not as specified
		Separation of color or finish rough (i.e., lint, dust, dirt, or other foreign matter embedded in finish)
		Finish not continuous, smooth, and adherent (i.e., orange peel, wrinkles, drops, streaks)
	Zinc, chromium, and cadmium plated	Coating omitted or area of no coating
		Coating not smooth, uniform, fine-grained, or not tightly adherent (i.e., flaky, blistered, or peeling); or coating is scratched, stained, discolored, shaded, broken, or cut through or is not free from pits, modules, or indications of burning
	Plastic	Not smooth, lusterless, or semi-gloss black
		Gloss exceeds SAE-AMS-STD-595/27038 when compared visually
	Polished finish	Polish not bright
		Buff drag, cloudy, hazy, or mottle, finish clearly noticeable

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TABLE II. End item visual defects - Continued.

Examination	Defect
Construction and workmanship	Sharp edge or projection that may cause injury
	Cracked, broken, bent, malformed, or damaged
	Pronounced tool or forming marks
Casting	Miscast, incomplete, or contains blowholes or evidence of porosity
Roller (when applicable)	Does not turn freely
	Joint not butted
Workmanship of welds or brazing	Weld or brazing missing, unless welding is not required
	Completely broken or not fused
	Cracked, partially broken, incomplete, or evidence of poor fusion or burn through
	Evidence that weld contains gas pockets or has been undercut or that excess flux or spatter has not been removed

4.3 Conformance inspection. Unless otherwise specified herein, sampling for conformance inspection shall be performed in accordance with MIL-STD-1916 or ASQ H1331.

4.3.1 Lot definitions. The lot definition is directly related to the stage of manufacturing at which a sample is taken.

4.3.1.1 Material composition test for copper-nickel-zinc (nickel silver) alloy. For purposes of sampling, a lot shall consist of material taken from one manufacturing run. Each test sample will be 4-ounces (oz). The lot size will be determined by dividing the weight of the lot by 4 oz. Example: a 100 pound lot will be considered to have a lot size of 400.

4.3.1.2 In-process visual examination. The lot size shall be the number of in-process rings of one configuration, class, and size, manufactured using the same materials and processes. The lot shall be unfinished rings without rollers.

4.3.1.3 End unit inspection. The lot size shall be the number of finished rings of one configuration, class, size, and style if applicable, manufactured using the same materials and processes.

4.3.2 Sampling plan. Unless otherwise specified, sampling procedures for conformance inspections and examinations shall be in accordance with 4.3.1 and MIL-STD-1916 or ASQ H1331. The acceptance quality limits (AQL) or verification levels (VL) shall be specified in the contract (see 6.2). Unless otherwise specified in the contract, sampling procedures and conformance inspections shall be in accordance with MIL-STD-1916, VL II or ASQ H1331, AQL 4.0.

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4.3.2.1 Sampling for material composition test for copper-nickel-zinc (nickel silver) alloy. A random sample of 4 oz test samples shall be selected from each lot and subjected to the inspection specified in 4.3.3.

4.3.2.2 Sampling for in-process visual examination. A random sample of in-process rings shall be selected from each lot and subjected to the visual examination specified in 4.3.4.2.

4.3.2.3 Sampling for end unit inspection. A random sample of rings shall be selected from each lot and subjected to the inspections specified in 4.3.5.1.

4.3.3 Component and material inspection. Unless otherwise specified herein or in the acquisition order (see 6.2), components and materials shall be inspected in accordance with all the requirements of the referenced documents. In addition, copper-nickel-zinc (nickel silver) alloy shall conform to the chemical composition of 3.2.6, and shall be tested in accordance with ASTM E1282.

4.3.3.1 Inspection of nylon plastic material. Each batch of reground material, when applicable, shall be inspected for compliance with the requirements specified in 3.2.8. Any batch containing reground nylon plastic material exceeding 25 percent of a blended mixture with virgin nylon material shall be cause for rejection of the batch.

4.3.4 In-process examinations.

4.3.4.1 In-process examination for zinc-plating. Samples of in-process class 2 and class 3 rings, except configurations L and X taken in 4.3.2.2 shall be examined to confirm that the phosphate treated zinc-plating is applied before enameling as specified in 3.4.2.2.

4.3.4.2 In-process visual examination of welded or brazed rings without roller. Samples of welded or brazed rings taken in 4.3.2.2 shall be examined before finishing and assembling the roller to determine conformance with 3.3.1 or 3.3.2.

4.3.5 End item examinations.

4.3.5.1 End item visual and dimensional examination. Samples of the end items taken in 4.3.2.3 shall be examined for the defects listed in table II, and for compliance with all dimensions specified in 3.3.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the inventory control point's packaging activities within the military service or defense agency, or within the military service's system command. Packaging data retrieval is available from the managing military department's or defense agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use.

6.1.1 Class 1 rings. Class 1 rings, other than configurations W, X, and L, are intended for use in conjunction with textile, webbing, and leather items where the corrosion products of ferrous metals may adversely affect the textile, webbing, and leather items.

6.1.2 Classes 2 and 3 rings. Classes 2 and 3 rings, other than configurations X and D, are intended for use as alternates to class 1 rings in periods of emergency. These also may be used where a specific requirement for additional strength is necessary.

6.1.3 Class 4 rings. Class 4 rings are intended for use on animal equipment in special applications where high resistance to corrosion is required.

6.1.4 Configurations W, X, and L rings. Configuration W rings are generally used on munitions command items; configuration X rings are generally used on weapons command items; configuration L rings are generally used on electronic command items.

6.1.5 Configuration D rings. Configuration D rings are generally used on animal equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Class required (see 1.2 and 1.2.1).
c. Configuration required (see 1.2 and 1.2.2).
d. Size required, when applicable (see 1.2 and 1.2.3).
e. Style required, when applicable (see 1.2 and 1.2.4).
f. First article inspection, when required (see 3.1 and 4.2).
g. When configuration "A" ring should be welded (see 3.3.2).
h. Finish required (see 3.4.5.1 and 3.4.6.2).
i. AQL/VL (see 4.3.2).
j. Component and material conformance inspection requirements, if different (see 4.3.3).
k. Packaging (see 5.1).

6.3 Safety Data Sheet (SDS). If the use of cadmium is not considered an article as specified in OSHA 3844, “Hazard Communication: Hazardous Classification Guidance for Manufacturers, Importers, and Employers”, a SDS must be prepared and submitted in accordance with FED-STD-313 (obtain from https://assist.dla.mil). The SDS must also meet the requirements of 29 CFR 1910.1200. The 29 CFR 1910.1200 requires that the SDS for each hazardous chemical used in an operation must be readily available to personnel using the material.

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6.4 Subject term (key word) listing.

Animal equipment
Cadmium
Equipage
Harness
Leather
Nylon
Rollers
Strapping
Tack
Textile
Webbing

6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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Navy - MC
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DLA - GS4

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